Data frame functions

Warmup activity

Work on the activity (handout) with a neighbor, then we will discuss as a class

Warmup

```
1 grouped_max <- function(df, group_var, max_var) {
2   df |>
3     group_by(group_var) |>
4     summarize(max(max_var, na.rm=T))
5 }
6
7 grouped_max(penguins, species, bill_depth_mm)
```

What is this code *trying* to do?

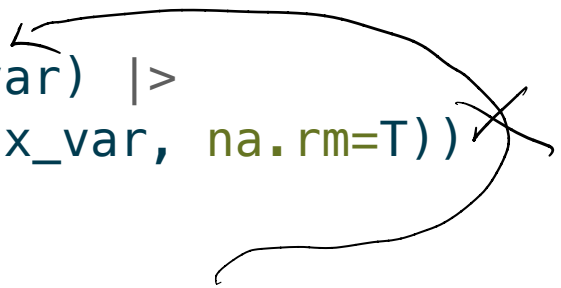
Hoping:

```
penguins |>
  group_by(species) |>
  summarize(max(bill_depth_mm, na.rm=T))
```

Warmup

```r
grouped_max <- function(df, group_var, max_var) {
  df |>
    group_by(group_var) |>
    summarize(max(max_var, na.rm=T))
}

grouped_max(penguins, species, bill_depth_mm)
```



(not happening)

```
Error in `group_by()`:
! Must group by variables found in `.data`.
✖ Column `group_var` is not found.
```

What is causing the error?

what R is actually evaluating:

```
penguins |>
  group_by(group_var) |>
  ...
```

problem: group_var is not a column in penguins data

Warmup

```
1 grouped_max <- function(df, group_var, max_var) {
2   df |>
3     group_by(group_var) |>
4     summarize(max(max_var, na.rm=T))
5 }
6
7 grouped_max(penguins, species, bill_depth_mm)
```

```
Error in `group_by()`:
! Must group by variables found in `.data`.
✖ Column `group_var` is not found.
```

What should we change so the code runs correctly?

Embracing

```
1 grouped_max <- function(df, group_var, max_var) {
2   df |>
3     group_by({{ group_var }}) |>
4     summarize(max({{ max_var }}, na.rm=T))
5 }
6
7 grouped_max(penguins, species, bill_depth_mm)
```

don't look for a column that is literally called "group_var"

instead, look for a column w/ name matching input for group_var in function

```
# A tibble: 3 × 2
  species   `max(bill_depth_mm, na.rm = T)`
  <fct>                               <dbl>
1 Adelie                               21.5
2 Chinstrap                            20.8
3 Gentoo                               17.3
```

penguins |>
  group_by(species) |>
  ...

Why do we need embracing?

```
1 penguins |>
2   filter(species == "Adelie")
```

This code contains two different types of variables:

- `penguins` is an **env-variable** (environment variable)
- `species` is a **data-variable** (it makes sense only within the context of a data frame)

Env-variables

Env-variables are objects in the R environment that we can interact with directly. For example:

```
1 head(penguins)
```

```
# A tibble: 6 × 8
  species island    bill_length_mm bill_depth_mm flipper_length_mm
body_mass_g
  <fct>   <fct>              <dbl>         <dbl>             <int>
<int>
1 Adelie  Torgersen           39.1          18.7               181
3750
2 Adelie  Torgersen           39.5          17.4               186
3800
3 Adelie  Torgersen           40.3          18                 195
3250
4 Adelie  Torgersen           NA            NA                  NA
NA
```

Data-variables

Data-variables only exist in the context of a data frame:

```
1 # R doesn't know what 'species' is:
2 species
```

```
Error: object 'species' not found
```

```
1 # R DOES understand species in the context of penguins:
2 penguins$species
```

```
  [1] Adelie    Adelie    Adelie    Adelie    Adelie    Adelie
Adelie
  [8] Adelie    Adelie    Adelie    Adelie    Adelie    Adelie
Adelie
 [15] Adelie    Adelie    Adelie    Adelie    Adelie    Adelie
Adelie
 [22] Adelie    Adelie    Adelie    Adelie    Adelie    Adelie
Adelie
 [29] Adelie    Adelie    Adelie    Adelie    Adelie    Adelie
Adelie
 [36] Adelie    Adelie    Adelie    Adelie    Adelie    Adelie
Adelie
```

Tidy evaluation

Many `tidyverse` functions are nice and allow us to reference *data-variables*:

```
penguins |>
  filter(species == "Adelie")
```

Here `filter` **knows** to look for a column called `species` in the `penguins` data.

Tidy evaluation

Of course, you will get an error if you try to reference a data-variable that doesn't exist! E.g. if we mis-spell the name:

```
1 penguins |>
2   filter(speices == "Adelie")
```

```
Error in `filter()`:
ℹ In argument: `speices == "Adelie"`.
Caused by error:
! object 'speices' not found
```

Tidy evaluation

Of course, you will get an error if you try to reference a data-variable that doesn't exist!

```
penguins |>
  group_by(group_var) |>
  summarize(max(max_var, na.rm=T))
```

```
Error in `group_by()`:
! Must group by variables found in `.data`.
✖ Column `group_var` is not found.
```

The problem: `group_var` and `max_var` are not columns in the `penguins` data!

Tidy evaluation

```r
grouped_max <- function(df, group_var, max_var) {
  df |>
    group_by(group_var) |>
    summarize(max(max_var, na.rm=T))
}

grouped_max(penguins, species, bill_depth_mm)
```

```
Error in `group_by()`:
! Must group by variables found in `.data`.
✖ Column `group_var` is not found.
```

What we **want** R to run:

```r
penguins |>
  group_by(species) |>
  summarize(max(bill_depth_mm, na.rm=T))
```

```
# A tibble: 3 × 2
  species   `max(bill_depth_mm, na.rm = T)`
  <fct>                               <dbl>
1 Adelie                               21.5
```

2	Chinstrap	20.8
3	Gentoo	17.3

Tidy evaluation

```r
grouped_max <- function(df, group_var, max_var) {
  df |>
    group_by(group_var) |>
    summarize(max(max_var, na.rm=T))
}

grouped_max(penguins, species, bill_depth_mm)
```

```
Error in `group_by()`:
! Must group by variables found in `.data`.
✖ Column `group_var` is not found.
```

What R is **actually** running:

```r
penguins |>
  group_by(group_var) |>
  summarize(max(max_var, na.rm=T))
```

```
Error in `group_by()`:
! Must group by variables found in `.data`.
✖ Column `group_var` is not found.
```

The solution: embracing

```r
grouped_max <- function(df, group_var, max_var) {
  df |>
    group_by({{ group_var }}) |>
    summarize(max({{ max_var }}, na.rm=T))
}

grouped_max(penguins, species, bill_depth_mm)
```

```
# A tibble: 3 × 2
  species   `max(bill_depth_mm, na.rm = T)`
  <fct>                               <dbl>
1 Adelie                               21.5
2 Chinstrap                            20.8
3 Gentoo                               17.3
```

What R is running now:

```r
penguins |>
  group_by(species) |>
  summarize(max(bill_depth_mm, na.rm=T))
```

```
# A tibble: 3 × 2
  species   `max(bill_depth_mm, na.rm = T)`
  <fct>                               <dbl>
1 Adelie                               21.5
2 Chinstrap                            20.8
3 Gentoo                               17.3
```

Another example

Suppose we want to fit a simple linear regression model:

```
penguins |>
  lm(bill_length_mm ~ bill_depth_mm, data = _) |>
  coef()
```

```
(Intercept) bill_depth_mm
 55.0673698    -0.6498356
```

Another example

```
penguins |>
  lm(bill_length_mm ~ bill_depth_mm, data = _) |>
  coef()
```

```
(Intercept) bill_depth_mm
 55.0673698    -0.6498356
```

```
lm_coef <- function(df, x, y) {
  df |>
    lm({{ y }} ~ {{ x }}, data = _) |>
    coef()
}

lm_coef(penguins, bill_depth_mm, bill_length_mm)
```

Do you think this code will work?

Another example

```
penguins |>
  lm(bill_length_mm ~ bill_depth_mm, data = _) |>
  coef()
```

```
(Intercept) bill_depth_mm
 55.0673698    -0.6498356
```

```
lm_coef <- function(df, x, y) {
  df |>
    lm({{ y }} ~ {{ x }}, data = _) |>
    coef()
}

lm_coef(penguins, bill_depth_mm, bill_length_mm)
```

```
Error: object 'bill_length_mm' not found
```

Why does this code fail?

Another example

```
penguins |>
  lm(bill_length_mm ~ bill_depth_mm, data = _) |>
  coef()
```

```
(Intercept) bill_depth_mm
 55.0673698    -0.6498356
```

```
lm_coef <- function(df, x, y) {
  df |>
    lm({{ y }} ~ {{ x }}, data = _) |>
    coef()
}

lm_coef(penguins, bill_depth_mm, bill_length_mm)
```

```
Error: object 'bill_length_mm' not found
```

Problem: The `lm` function does not support tidy evaluation!

(To see if a function does support tidy evaluation, look for

data-masking
tidy-selection

(key words in documentation)

Fixing the issue

```
penguins |>
  lm(bill_length_mm ~ bill_depth_mm, data = _) |>
  coef()
```

```
(Intercept) bill_depth_mm
 55.0673698    -0.6498356
```

```
lm_coef <- function(df, x, y) {
  df |>
    lm({{ y }} ~ {{ x }}, data = _) |>
    coef()
}

lm_coef(penguins, bill_depth_mm, bill_length_mm)
```

```
Error: object 'bill_length_mm' not found
```

If `lm` doesn't support tidy evaluation, what could we do differently?

Fixing the issue

SLR slope: $\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$

```
penguins |>
  lm(bill_length_mm ~ bill_depth_mm, data = _) |>
  coef()
```

```
  (Intercept) bill_depth_mm
   55.0673698    -0.6498356
```

```
penguins |>
  summarize(slope = cov(bill_depth_mm, bill_length_mm,
                        use="complete.obs")/
              var(bill_depth_mm, na.rm=T))
```

```
[1] -0.6498356
```

Fixing the issue

```
penguins |>
  summarize(slope = cov(bill_depth_mm, bill_length_mm,
                        use="complete.obs")/
              var(bill_depth_mm, na.rm=T))
```

How would I turn this into a function?

```
slr_slope <- function(df, x, y) {
  df |>
    summarize(slope = cov({{x}}, {{y}}, ...) /
                var({{x}}, ...))
}
```

Fixing the issue

```
slr_slope <- function(df, x, y) {
  df |>
    summarize(slope = cov({{ x }}, {{ y }}, use="complete.obs")/
                  var({{ x }}, na.rm=T))
}

slr_slope(penguins, bill_depth_mm, bill_length_mm)
```

```
# A tibble: 1 × 1
   slope
   <dbl>
1 -0.650
```

```
slr_slope(penguins, flipper_length_mm, bill_length_mm)
```

```
# A tibble: 1 × 1
  slope
  <dbl>
1 0.255
```

Class activity

Work on the class activity (course website).